Name

Class

R. No.

- Mathematics
- IIT JEE

- Test 2 (July 13 , 2010)
- Absolute Value of a Real Number

- Maximum Marks 30
- Time 30 minutes

Instructions

1. **The test contains two subjective questions(Q1 and Q2) and two write the answer type questions(Q3, Q4).**
2. **The marking scheme will include marks for partially correct answers also.**
3. **For the questions of the 'write the answer' type, only the answer needs to be written in the specified space. Separate space has been provided for rough work.**

Q1: Write the complete solution set of $|[x]| = [|x|]$ 8 Marks

Q2: The Complete Solution set of $2|\tan x| = \tan x + \sqrt{3}$ is

6 Marks

Comprehension

A function $f_n(x)$ is defined for all $n \in N$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \frac{2x-1}{x+1}$ for $x \in R - \{-1\}$. Using this definition , $f_2(x) = f_1(f_1(x)) = \frac{x-1}{x}$ for $x \in R - \{-1,0\}$. Similarly , $f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x}$ for $x \in R - \{-1,0,\frac{1}{2}\}$ and so on. Based on the above information, answer the questions below.

Q3: The complete solution set of $|f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$ 8 Marks

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Q4: The complete solution set of x for which $|f_{56}(x)| > f_{56}(x)$ 8 Marks

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Space for Rough Work